

Matplotlib Charts - Syntax with Examples

Common Import:

```
import matplotlib.pyplot as plt
```

Line Chart

Syntax:

```
plt.plot(x, y)
```

Example:

```
x=[1, 2, 3, 4]
y=[10, 15, 12, 20]
plt.plot(x, y, marker='o')
plt.show()
```

Bar Chart

Syntax:

```
plt.bar(x, y)
```

Example:

```
students=['A', 'B', 'C']
marks=[80, 90, 75]
plt.bar(students, marks)
plt.show()
```

Horizontal Bar

Syntax:

```
plt.barh(x, y)
```

Example:

```
plt.barh(students, marks)
plt.show()
```

Histogram

Syntax:

```
plt.hist(data, bins=5)
```

Example:

```
data=[10, 12, 15, 18, 20, 21, 22, 24, 25]
plt.hist(data, bins=5)
plt.show()
```

Scatter Plot

Syntax:

```
plt.scatter(x, y)
```

Example:

```
x=[1, 2, 3, 4]
y=[5, 8, 6, 10]
plt.scatter(x, y)
plt.show()
```

Pie Chart

Syntax:

```
plt.pie(values, labels=labels, autopct='%1.1f%%')
```

Example:

```
labels=['A', 'B', 'C']
values=[40, 35, 25]
plt.pie(values, labels=labels, autopct='%1.1f%%')
plt.show()
```

Box Plot

Syntax:

```
plt.boxplot(data)
```

Example:

```
data=[[10, 20, 30, 40, 100]]
plt.boxplot(data)
plt.show()
```

Violin Plot

Syntax:

```
plt.violinplot(data)
```

Example:

```
data=[[10, 20, 30, 40, 50]]
plt.violinplot(data)
plt.show()
```

Area Chart

Syntax:

```
plt.fill_between(x,y)
```

Example:

```
x=[1, 2, 3, 4]
y=[2, 4, 3, 5]
plt.fill_between(x, y)
plt.show()
```

Stack Plot

Syntax:

```
plt.stackplot(x,y1,y2)
```

Example:

```
x=[1, 2, 3]
y1=[2, 3, 4]
y2=[1, 2, 3]
plt.stackplot(x, y1, y2)
plt.show()
```

Stem Plot

Syntax:

```
plt.stem(x,y)
```

Example:

```
x=[1, 2, 3]  
y=[2, 5, 3]  
plt.stem(x,y)  
plt.show()
```

Step Chart

Syntax:

```
plt.step(x,y)
```

Example:

```
plt.step([1, 2, 3, 4], [5, 8, 6, 9])  
plt.show()
```

Error Bar

Syntax:

```
plt.errorbar(x,y,yerr=e)
```

Example:

```
x=[1, 2, 3]  
y=[5, 6, 7]  
e=[0.5, 0.3, 0.4]  
plt.errorbar(x,y,yerr=e)  
plt.show()
```

Hexbin

Syntax:

```
plt.hexbin(x,y)
```

Example:

```
plt.hexbin(x,y,gridsize=10)  
plt.show()
```

Event Plot

Syntax:

```
plt.eventplot(data)
```

Example:

```
data=[[1, 2, 3], [2, 4, 6]]  
plt.eventplot(data)  
plt.show()
```

Polar Plot

Syntax:

```
plt.polar(theta,r)
```

Example:

```
import numpy as np  
t=np.linspace(0, 6.28, 100)
```

```
r=t  
plt.polar(t,r)  
plt.show()
```

Contour Plot

Syntax:

```
plt.contour(X,Y,Z)
```

Example:

```
plt.contour(X,Y,Z)  
plt.show()
```

Filled Contour

Syntax:

```
plt.contourf(X,Y,Z)
```

Example:

```
plt.contourf(X,Y,Z)  
plt.show()
```

Image Plot

Syntax:

```
plt.imshow(img)
```

Example:

```
plt.imshow(img)  
plt.show()
```

Heatmap

Syntax:

```
plt.imshow(data); plt.colorbar()
```

Example:

```
data=[[1,2],[3,4]]  
plt.imshow(data)  
plt.colorbar()  
plt.show()
```

3D Surface

Syntax:

```
ax.plot_surface(X,Y,Z)
```

Example:

```
from mpl_toolkits.mplot3d import Axes3D  
fig=plt.figure()  
ax=fig.add_subplot(111,projection='3d')  
ax.plot_surface(X,Y,Z)
```

Frequently Used Functions:

```
plt.title(), plt.xlabel(), plt.ylabel(), plt.legend(), plt.grid(), plt.xlim(), plt.ylim(), plt.xticks(), plt.yticks(),  
plt.figure(), plt.subplot(), plt.tight_layout(), plt.savefig(), plt.show()
```